Math 242, S13
Quiz 10

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Find Laplace transforms of the following functions:
(a) $f(t) = -2e^{3t}$

$$\mathcal{L}\{f(t)\} = -2\mathcal{L}\{e^{3t}\} = -2\cdot\frac{1}{s-3}$$
$$= -\frac{2}{s-3}$$

(b) $f(t) = 2\sin 3t + \cos 3t$

$$\mathcal{L}\{f(t)\} = 2\mathcal{L}\{\sin 3t\} + \mathcal{L}\{\cos 3t\}$$
$$= 2\cdot\frac{3}{s^2+3^2} + \frac{s}{s^2+3^2}$$
$$= \frac{6+s}{s^2+9}$$

Problem 2. (5 points) Find inverse Laplace transforms of the following functions:
(a) $F(s) = \dfrac{6}{s^4}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{6}{s^4}\right\} = t^3$$

(b) $F(s) = \dfrac{8}{s^2-4}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{8}{s^2-4}\right\} = 4\mathcal{L}^{-1}\left\{\frac{2}{s^2-2^2}\right\}$$
$$= 4\sinh 2t$$